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▼ Welcome to TinyML

Litterature

Hardware

Software

 Lesson plan fall 2025

▶ Part 1 | Introduction to TinyML & ML Basics

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▶ Part 3 | Data Acquisition

▶ Part 3.1 | Build a data acquisition system

▶ Part 5 | Feature engineering

▶ Part 5.1 | Implement feature calculation on your devic

Part 6 | Introduction to Machine Learning

▶ Part 7 | Introduction to EM Learn, ML on Microcontr

Part 7.1 | Implement a ML model trained on your fei

▶ Part 8 | Introduction to Deep Learning

▶ Part 8.1 | Introduction to Convolutions

▶ Part 9 | Introduction to AiFES, Deep Learning on

Part 10 | Optimisation techniques, Quantisation

Part 11 | Optimisation techniques, Pruning

▶ Part 12 | Anomaly detection

▶ Semester/Exam Project

Litterature

In this course we will be using primarily open litterature in the form of an online book, tool manuals & examples and a number of papers.

A lot of litterature on Machine Learning is available, and if you are familiar with the topic, you may already have books or pdf's available.

Course Litterature

Machine Learning Systems:
Machine learning, and TinyML in particular is a vibrant field that evolves continously, will be using chapters from this book, that is being written and updated as we speak, fortunately its open acess, and you can access it freely.

FRONT MATTER

Preface

Dedication

Acknowledgements

Contributors

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About the Book

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REFERENCES

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LABS

Setup Nicla Vision

CV on Nicla Vision

Object Detection

Audio Feature Engineering

Keyword Spotting (KWS)

DSP - Spectral Features

Motion Classification and Anomaly Detection

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Machine Learning Systems

with TinyML

ABSTRACT

Machine Learning Systems with TinyML offers readers an entry point to understand comprehensive machine learning systems by grounding concepts in accessible TinyML applications. As resource-constrained edge computing sees rapid expansion, the ability to construct efficient ML pipelines grows crucial. This book aims to demystify the process of developing complete ML systems suitable for deployment - spanning key phases like data collection, model design, optimization, acceleration, security hardening, and integration. The text touches on the full breadth of concepts relevant to general ML engineering across industries and applications through the lens of TinyML. Readers will learn basic principles around designing ML model architectures, hardware-aware training strategies, performant inference optimization, benchmarking methodologies and more. Additionally, crucial systems considerations in areas like reliability, privacy, responsible AI, and solution validation are also explored in depth. In summary, the book strives to equip newcomers and professionals alike with integrated knowledge covering full stack ML system development, using easily accessible TinyML applications as the vehicle to impart universal concepts required to unlock production ML.

Preface

Welcome to Machine Learning Systems with TinyML. This book is your gateway to the fast-paced world of AI systems through the lens of embedded systems. It is an extension of the course, TinyML from [CS249r](#) at Harvard University.

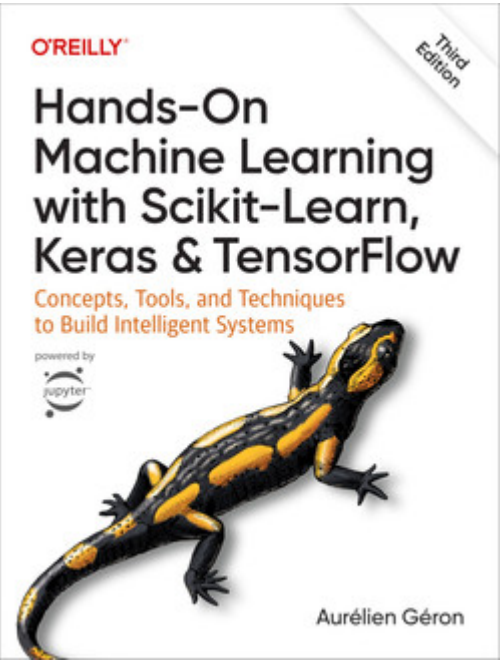
Our aim is to make this open-source book a collaborative effort that brings together insights from students, professionals, and the broader community of applied machine learning practitioners. We want to create a one-stop guide that dives deep into the nuts and bolts of AI systems and their many uses.

"If you want to go fast, go alone. If you want to go far, go together." – African Proverb

This isn't just a static textbook; it's a living, breathing document. We're making it open-source and continually updated to meet the ever-changing needs of this dynamic field. Expect a rich blend of expert knowledge that guides you through the complex interplay between cutting-edge algorithms and the foundational principles that make them work. We're setting the stage for the next big leap in tech innovation.



Hands-on Machine Learning with Scvikit-Learn, Keras & TensorFlow:
In addition, i highly recommend this excellent book "Hands-On Machine Learnin with Scikit-Learn, Keras and TensorFlow, to get familiar with popular python libraries, SciKit-Learn and Keras, as well as machine learning in general. This book will take you much further in Machine Learning and Deep Learning than the topics we cover in the introductory parts of this course.



The book has an open accompanying github repository, with a lot of excellent code examples.